

When Rankings Reward Age: Admissions Formula and the College Application Gap

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April 25, 2026

[Preliminary draft, do not cite]

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College admissions formulas determine which student characteristics translate into admission chances, and when they change, they can interact with differences along dimensions the designer did not target. I study Chile's 2013 ranking reform, which required programs to place at least 10 percent of admissions weight on a student's class rank within their own school. Because age at school entry affects academic performance, the reform raised the weight on a dimension correlated with the school-entry cutoff. Using a difference-in-discontinuities design, I find that the reform widened the application gap at the July 1 cutoff by 3.8 percentage points (12 percent) and the GPA gap by 0.082 standard deviations. The response concentrates in secondary-educated, higher-income families. Marginal applicants are less likely to be admitted or to enroll, producing no net gain in enrollment. Formula design can interact with age-at-entry differences to produce distributional effects in college access.

Keywords: College admissions, admissions formula design, application behavior, education policy

JEL: I21, I23, I24, I28

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1 Introduction

Two students sit in the same classroom, take the same courses, and graduate from the same school. They are, in almost every observable dimension, identical. Yet because they were born two days apart, one on each side of a school entry cutoff, one student is about a year older than the other throughout schooling. By twelfth grade, the older student tends to earn slightly higher grades than the younger one, a pattern the literature attributes to a mix of channels that includes accumulated skills, parental beliefs and investment, and persistent differences in early learning (Bedard and Dhuey, 2006; Celhay and Gallegos, 2025; Elder and Lubotsky, 2009; McEwan and Shapiro, 2008). Whether this advantage matters for college access depends on how the admissions formula is designed. A formula that rewards absolute performance treats both students' grades the same way. A formula that rewards relative position within the school amplifies the gap, because the older student ranks higher among the very peers whose grades reflect the same curriculum and the same teachers. The difference is not in who has higher grades. It is in what those grades buy.

This paper studies whether admissions formula design can amplify pre-existing age-at-entry differences in academic performance into differences in college access. A large literature has established that age at school entry affects grades, test scores, and long-run educational attainment (Angrist and Krueger, 1992; Bedard and Dhuey, 2006; Black et al., 2011; Celhay and Gallegos, 2025; Elder and Lubotsky, 2009; Fredriksson and Öckert, 2014; McEwan and Shapiro, 2008), with this advantage persisting into college outcomes (Matta et al., 2016). Less is known about how institutional features of the college admissions process mediate these effects downstream. Admissions formulas translate academic performance into admission chances, and the specific way they do so determines which student characteristics are rewarded. If a formula change increases the weight on a dimension along which older students differ from their younger peers, the formula can interact with that difference even if the designer intended no such effect.

Chile's 2013 ranking reform provides a direct test of this hypothesis. Before the reform, college admissions scores were computed from absolute GPA and standardized entrance exam scores, both compared against the national distribution of applicants. In 2013, the Council of Rectors introduced a within-school class ranking component (*puntaje ranking*), requiring programs to assign at least 10

percent of their admissions weight to a student’s relative position in the GPA distribution of their own school. In Chile specifically, [McEwan and Shapiro \(2008\)](#) document that delayed enrollment around this cutoff raises fourth- and eighth-grade test scores, and [Celhay and Gallegos \(2025\)](#) show that early skill advantages translate into higher PSU scores and college enrollment through parental beliefs and investment. The reform raised the admissions weight placed on within-school relative performance, a dimension plausibly correlated with age at school entry through these documented channels. I exploit the timing of this reform using a difference-in-discontinuities design ([Grembi et al., 2016](#)), comparing the magnitude of the birthday discontinuity in college outcomes before and after 2013 at the July 1 school entry cutoff.

The reform produced measurable responses at the July 1 cutoff. After the reform, older students near the cutoff earned significantly higher grades relative to their younger peers ($\hat{\delta} = 0.082$, $p = 0.037$), consistent with the ranking score increasing the return to within-school performance. This translated into a 3.8 percentage point increase in the application gap at the cutoff, equivalent to 12 percent of the pre-reform application rate. These additional applications, however, did not produce additional enrollments. Conditional enrollment among applicants fell by 7.0 percentage points, with patterns consistent with marginal applicants who aim higher than their competitive position warrants and are more likely to pursue alternatives when admitted. The net effect on unconditional enrollment is zero. The reform expanded who applies without expanding who enrolls.

The response is not uniform across families. Students from secondary-educated, higher-income households, with sufficient resources and information to act on a new signal, exhibit the largest response ($\hat{\delta} = 0.113$, $p = 0.006$). Before the reform, older students in this group were no more likely to apply than their younger peers. After the reform, they were substantially more likely to do so. College-educated families show no response because their children apply regardless. Primary-educated families show no response because the ranking signal alone does not overcome the larger barriers they face. The reform matters most at the margin of college participation.

This paper contributes to several literatures. A growing body of work studies the consequences of admissions formulas for equity and sorting ([Concha-Arriagada, 2023](#); [Cullen et al., 2013](#); [Larroucau et al., 2015](#); [Niu and Tienda, 2010](#)), focusing on how formulas redistribute access across schools or socioeconomic groups. This paper shows that formula changes can also interact with characteristics the

designer did not intend to reward, producing differential effects along a demographic dimension rather than the intended sorting. The finding connects two literatures that have developed independently. The age-at-entry literature (Bedard and Dhuey, 2006; Black et al., 2011; Celhay and Gallegos, 2025; Fredriksson and Öckert, 2014; McEwan and Shapiro, 2008) documents that age at school entry affects academic outcomes through multiple channels, including accumulated skills, parental investment, curriculum placement, and peer-rank effects, but has not examined how admissions institutions mediate these effects downstream. The ordinal rank literature (Denning et al., 2023; Elsner and Isphording, 2017; Murphy and Weinhardt, 2020) shows that within-school rank shapes educational investment and long-run outcomes, but has studied rank primarily as an informal psychological mechanism operating through peer comparison and self-concept. Chile’s reform formalizes rank into a policy instrument, and the results show that this formalization creates distributional consequences that the informal channel does not. Finally, the three-margin decomposition of application, admission, and enrollment reveals that a reform can expand who applies to college without expanding who enrolls. A growing literature on college application behavior documents that small changes in costs or information shift who applies (Hoxby and Avery, 2012; Pallais, 2015), but less attention has been paid to what happens to marginal applicants after they enter the admissions process. The enrollment decomposition highlights these downstream barriers.

The remainder of the paper proceeds as follows. Section 2 describes Chile’s school entry rules, the college admissions system, and the 2013 ranking reform. Section 3 presents the data sources and sample construction. Section 4 develops the difference-in-discontinuities design. Section 5 reports the main results, including covariate balance, the enrollment decomposition, and socioeconomic heterogeneity. Section 6 concludes.

2 Institutional Background

2.1 School Entry Rules

Chile operates a mandatory education system spanning first grade through the completion of secondary school (grade 12). Students enroll in *educación básica* (grades 1 through 8) and then *educación media*

(grades 9 through 12). Children must turn six years old on or before April 1 to begin first grade that academic year. This rule implies a minimum enrollment age of approximately 5.92 years, calculated as of March 1st of the corresponding year.

Until 2017, however, a Ministry of Education decree allowed schools to adopt later cutoff dates, as late as July 1.¹ In practice, most schools in Chile used July 1 as their operational cutoff, with smaller shares adopting May 1 or June 1 (McEwan and Shapiro, 2008). A child born on July 2 would need to wait an additional year to begin first grade, entering school at approximately age 6.67 rather than 5.67 by March 1st of the corresponding year. This creates a discontinuity of up to one year in school starting age at each cutoff date, with the largest and most widely used discontinuity occurring on July 1. In practice, the observed discontinuity is approximately 0.4 years (Figure 1) because some parents of children born before July 1 choose to delay enrollment voluntarily, narrowing the gap relative to those born after the cutoff who are required to wait. This partial compliance motivates the use of intention-to-treat estimates throughout the paper rather than scaling by enrollment age.

Two features of the Chilean setting make it particularly useful for the present analysis. First, the July 1 cutoff generates the largest first-stage discontinuity in enrollment age among the four cutoffs (approximately 0.4 years), providing sufficient statistical power to detect differences in downstream outcomes such as college applications. Second, Chile’s Ministry of Education maintains comprehensive administrative records covering all schools nationally, and the college admissions process operates through a single centralized platform (DEMRE). Together, these allow us to track students throughout their entire educational history, from school enrollment to college application outcomes, using linked individual identifiers.

2.2 College Admissions

Admission to selective universities in Chile is managed through a centralized system coordinated by the Department of Evaluation, Measurement, and Educational Registry (DEMRE). Students can apply to up to nearly 2,000 programs hosted by 41 universities by submitting a rank-ordered list of up to ten preferred programs. Each program computes a composite admission score for every applicant as a

¹In 2017, Decree 1718 eliminated this flexibility and enforced April 1 as the sole enrollment cutoff for all schools.

weighted average of high school grade point average (NEM, *Notas de Enseñanza Media*), standardized entrance exam scores in mathematics and language, an optional test in either history or science, and, after 2013, a within-school class ranking score. Each program sets its own weights subject to regulatory floors established by the Council of Rectors (CRUCH), which require minimum weights on NEM, mathematics, language, and (after 2013) the ranking score.²

Formally, each program p assigns applicant i a composite admission score in year t as

$$S_{i,p,t} = w_{p,t}^{math} \cdot PSU_i^{math} + w_{p,t}^{lang} \cdot PSU_i^{lang} + w_{p,t}^{opt} \cdot PSU_i^{opt} + w_{p,t}^{nem} \cdot NEM_i + w_{p,t}^{rank} \cdot Ranking_{i,t}, \quad (1)$$

where $w_{p,t}^{rank} = 0$ for all programs in all years before 2013 and $w_{p,t}^{rank} \geq 0.10$ thereafter. The ranking score itself was undefined before 2013. The reform consisted of creating this component and requiring each program to assign it at least 10 percent of total weight. The remaining weights are set by each program subject to the regulatory floors for the other components.

The entrance exam, known as the *Prueba de Selección Universitaria* (PSU) during our study period, is taken by approximately 95 percent of secondary school graduates each year.³ Scores are scaled to a distribution with a range of 150 to 850 and a mean and median of 500. After receiving their scores, students submit their preference list, and the centralized system runs a student-proposing deferred acceptance algorithm (DAA) to assign students to programs (Hastings et al., 2013). The algorithm considers each student’s composite score for each listed program and matches them to their most preferred program in which their score exceeds the admission threshold. Each preference receives a disposition (selected, not selected, or superseded by a higher-ranked admission). We use these per-preference dispositions to examine application behavior in Section 5.3.

Once matched, students are expected to enroll in the assigned program. In practice, however, enrollment is not automatic. Some admitted students choose not to matriculate, opting instead for programs outside the centralized system, alternative educational paths, entry into the labor force, retaking the entrance exam in a later cycle, or other trajectories we do not fully observe. This

²The specific regulatory floors are 10 percent for NEM and 10 percent for the ranking score (post-2013). Mathematics and language must each receive positive weight, though the exact minimum varies by year. Every program must assign positive weight to at least one of the optional tests (history or science). See DEMRE’s annual admissions guidelines for details.

³The PSU was replaced by the *Prueba de Acceso a la Educación Superior* (PAES) starting in 2022, outside our sample period.

distinction between admission and enrollment becomes relevant when we examine the conditional enrollment margin in Section 5.3.

2.3 The 2013 Ranking Reform

Before 2013, all components of the admissions formula (NEM and standardized exam scores) were compared against the full national population of test-takers. A large literature documents age-at-cutoff effects on academic performance in other settings. For example, students who enter first grade at an older age tend to score higher on standardized achievement tests (Bedard and Dhuey, 2006; Elder and Lubotsky, 2009) and earn higher grades through secondary school (Black et al., 2011; McEwan and Shapiro, 2008). This body of evidence provides the conceptual backdrop against which a change in formula weights could produce differential effects at the July 1 cutoff, although the underlying channels may vary across settings and need not operate identically in Chile.

In 2013, the Council of Rectors introduced a new component in the admissions formula called the *puntaje ranking* (ranking score).⁴ The ranking score captures a student’s relative position within the GPA distribution of graduates from the same school, averaged over the three most recent graduating cohorts. Programs were required to assign at least 10 percent of their composite admissions weight to this new component.

The reform shifted the composition of what the admissions formula rewards. Programs funded the new ranking component by reducing the weight assigned to NEM, mathematics, and verbal exam scores.⁵ After the reform, the composite admissions score placed more weight on a student’s position within their own school and less weight on their position in the national distribution of NEM, mathematics, and verbal exam scores. A growing literature documents that within-school ordinal rank has independent effects on educational attainment, human capital investment, and long-run earnings (Denning et al., 2023; Elsner and Isphording, 2017; Murphy and Weinhardt, 2020), and the reform brought this dimension into the admissions formula for the first time.

⁴This change was established through Agreement 24/2012 (CRUCH). See Larroucau et al. (2015) for a detailed description of the reform and its implementation.

⁵On average, NEM weight fell by approximately 14 percentage points across programs, with smaller reductions in mathematics (3.6 pp) and verbal (3.6 pp) exam weights. See Appendix Table A.5 for the full decomposition.

Academic performance was already rewarded by the pre-reform formula through NEM and the standardized exam. The reform did not create a new pathway from student characteristics to admissions scores. It changed the relative weight of pathways already present, raising the return to within-school relative performance and lowering the return to components measured against the national distribution. This shift created a new incentive along the within-school margin. Any differential response at the July 1 cutoff could reflect a mechanical amplification of pre-existing academic differences, a behavioral response to the new incentive (such as differential effort or parental investment), or some combination. The present design cannot separate these mechanisms.

The identification strategy in this paper exploits the timing of this reform. If the reform produced a differential response at the July 1 age cutoff, then the discontinuity in college outcomes should increase after 2013 relative to the pre-reform period. The next sections describe the data and empirical framework used to test this prediction.

3 Data

3.1 Sources and Sample Construction

We combine student-level administrative records from two institutions within Chile’s Ministry of Education. The first source is the MINEDUC *Rendimiento* panel, which records every student’s enrollment, grade level, school, and academic outcomes (pass, fail, transfer) at the end of each academic year. From this panel we observe exact birth dates, gender, year of enrollment in each grade, and school identifiers.

The second source is the DEMRE database, which contains the universe of entrance exam (PSU) registrants from 2004 through 2020. For each test-taker, we observe scores in all four tests, high school GPA (NEM), the ranking score (available from 2013 onward), and whether the student submitted an application to the centralized admissions system. DEMRE also collects a short survey at the time of PSU registration, from which we construct measures of parental education (primary, secondary, or college) and family income (low, middle, or high).

We link the two sources using a unique identification number at the student level. Our estimation

sample consists of on-schedule, first-time PSU takers between 2010 and 2015 who attended urban, non-special-needs schools and have non-missing values for the key covariates.⁶ The resulting sample contains 1,158,950 students, roughly evenly split between the pre-reform period (2010–2012) and the post-reform period (2013–2015).

We focus on a symmetric three-year window around the 2013 reform, with the post-reform period ending in 2015 to avoid contamination from the 2016 *Gratuidad* reform (free tuition policy), which altered the college application margin through a separate channel.

The on-schedule restriction excludes approximately 4.9 percent of students (61,680 of 1,264,639) in PSU years 2010–2015 who took the entrance exam more than one year after secondary-school graduation. Most are one- or two-year delays, typically students who repeated a grade, took time between school and the exam, or retook it after a prior attempt. The restriction follows [Celhay and Gallegos \(2025\)](#) and is standard in the Chilean literature. It introduces a form of selection the present design cannot address. An alternative approach, defining reform exposure from birth date alone and measuring outcomes regardless of graduation-to-exam timing, would avoid this restriction and is left to future work.

3.2 Key Variables

Running variable. The running variable is the day of birth within the calendar year, normalized as the distance (in days) from the July 1 cutoff. Students born on or after July 1 (positive values) are “treated” in the sense that they are induced to delay first-grade enrollment by one year, entering school at an older age. Following [Celhay and Gallegos \(2025\)](#), we estimate intention-to-treat effects throughout and do not scale our estimates by enrollment age, as doing so would require monotonicity and exclusion restriction assumptions that are difficult to defend in this setting.⁷

⁶On-schedule means the student took the PSU in the same year of their secondary school graduation, in order to enroll into college the following year. We restrict our analysis to first-time takers to avoid confounding due to students who retake the exam after a gap year.

⁷Appendix A describes the construction of the enrollment age measure used in Figure 1, which combines directly observed first-grade enrollment records with an approximation following [McEwan and Shapiro \(2008\)](#) for the remainder of the sample.

Outcomes. Our primary outcome is *college application*, an indicator equal to one if the student submitted at least one application through the centralized admissions system after taking the PSU. We also study *college enrollment*, defined as admission to and enrollment in a program, both unconditionally (among all PSU takers) and conditionally (among applicants only). The three-margin decomposition of application, unconditional enrollment, and conditional enrollment is central to the paper’s argument. To investigate the enrollment margin further, we also construct indicators for whether an applicant was admitted to any listed program, whether they were admitted to their first-choice program, and whether they were admitted but chose not to enroll. These variables are derived from the per-preference disposition codes in the DEMRE application records.

GPA. We observe NEM (high school grade point average), normalized within each PSU cohort year to have mean zero and unit standard deviation. NEM captures the absolute academic performance channel that existed before the 2013 reform and serves as a mechanism check in the analysis.

Socioeconomic status. Parental education is classified into three mutually exclusive categories based on the highest level attained by either parent: primary only, secondary, and college. Family income is classified as low or high based on self-reported household income brackets from the DEMRE pre-test survey, where high combines the middle and upper brackets. These categories are predetermined relative to the student’s academic outcomes.

3.3 Descriptive Statistics

Table 1 presents summary statistics for students within a 15-day window on either side of the July 1 cutoff, separately by period (pre- and post-reform) and by position relative to the cutoff (younger and older at school entry).

Students born after the cutoff, who enter school at an older age, apply to college at higher rates than those born just before the cutoff in both periods. The gap, however, is noticeably larger after the reform (3.2 percentage points versus 2.3 percentage points pre-reform). This widening is consistent with a differential reform response at the cutoff. Whether this raw difference reflects a causal effect of the reform, rather than sustained trends in college access, is the question the empirical strategy

addresses.

Enrollment rates are also higher among students on the older side of the cutoff in both periods. Normalized GPA is slightly higher for these students than for those on the younger side (0.054 and 0.076 standard deviations in the pre- and post-reform periods, respectively), a direction consistent with age-at-cutoff effects on academic performance documented across settings (Bedard and Dhuey, 2006; McEwan and Shapiro, 2008).

Predetermined covariates such as gender, parental education, and family income appear similar across the cutoff in both periods, as expected under the assumption that birth dates near July 1 are quasi-randomly assigned. The sample is roughly balanced by gender (48 percent male), approximately half of the students have parents with secondary education, and about a quarter have parents with college education.

4 Empirical Strategy

4.1 Difference-in-Discontinuities Design

The school entry cutoff at July 1 creates a sharp discontinuity in first-grade enrollment age (Figure 1). Students born on or after July 1 must wait an additional year to begin first grade, entering school approximately 0.4 years older than students born just before the cutoff. As described in Section 2.3, the 2013 reform changed how the admissions formula weights within-school relative performance, creating a new incentive that can interact with the age difference at the cutoff.

To isolate the differential response produced by the 2013 ranking reform, we adopt a difference-in-discontinuities design following Grembi et al. (2016). The estimator compares the magnitude of the July 1 discontinuity before and after the reform:

$$\hat{\delta} = \hat{\beta}_{post} - \hat{\beta}_{pre} \quad (2)$$

where $\hat{\beta}_{pre}$ and $\hat{\beta}_{post}$ are separate RD estimates obtained from the pre-reform and post-reform subsamples, respectively. Because the two subsamples consist of non-overlapping cohorts, the standard

error is computed as:

$$SE(\hat{\delta}) = \sqrt{SE(\hat{\beta}_{pre})^2 + SE(\hat{\beta}_{post})^2} \quad (3)$$

Each period-specific estimate is obtained using the bias-corrected local polynomial estimator of [Calonico et al. \(2014\)](#) with a local linear polynomial ($p = 1$), a triangular kernel, and MSE-optimal bandwidth selection. We report bias-corrected point estimates with robust standard errors throughout, following [Cattaneo et al. \(2020\)](#).⁸

The estimand δ captures the change in the July 1 discontinuity attributable to the reform. A positive δ for college application would indicate that the reform widened the application gap between students on the older and younger sides of the cutoff.

4.2 Identification

The difference-in-discontinuities design rests on two assumptions ([Lee and Lemieux, 2010](#)). The first is the standard continuity condition for regression discontinuity designs: potential outcomes must be continuous at the July 1 cutoff. This would be violated if families systematically manipulate birth timing to place children on the advantaged side of the cutoff. The second is that no other factor changed the magnitude of the July 1 discontinuity at exactly 2013, other than the ranking reform ([Grembi et al., 2016](#)).

No manipulation of the running variable. Birth dates are determined well before parents could anticipate school entry cutoff rules with precision. Nevertheless, evidence from Chile shows a correlation between parental socioeconomic status and scheduled C-sections in private hospitals ([Borrescio-Higa and Valdés, 2019](#)), raising the possibility that higher-income parents may time births around cutoff dates. We examine the density of births around the July 1 cutoff using the test proposed by [McCrary \(2008\)](#) and find no visual evidence of bunching (Appendix Figure [A.1](#)). Moreover, the difference-in-discontinuities design provides additional protection against this concern. Any birth-timing manipulation would need to change differentially at 2013 to bias our estimates. Since the

⁸The conventional RD estimator has asymptotic bias. The bias-corrected estimator removes this bias using a pilot bandwidth, and the robust standard error accounts for the additional variance introduced by the correction. See [Calonico et al. \(2014\)](#) for details.

students in our sample were born approximately 17 to 18 years before the reform was announced, differential manipulation in response to the reform is not plausible.

No compositional changes at the threshold. A more direct test of the continuity assumption examines whether predetermined covariates jump discontinuously at the cutoff in a way that changed with the reform. We estimate the RD for each predetermined covariate separately in the pre- and post-reform periods, and compute the difference-in-discontinuities for each. The results of this exercise, presented in Section 5.2, confirm that no covariate exhibits a statistically significant differential change at the threshold.

Sample restriction and mechanism uncertainty. Two features of the sample and design warrant acknowledgment. First, the on-schedule restriction (Section 3.1) removes students who took the PSU more than one year after secondary-school graduation. This narrows the sample to students whose age difference at the cutoff reflects first-grade enrollment timing rather than grade repetition or strategic delay, but it introduces selection on unobserved factors correlated with on-schedule completion. Second, as discussed in Section 2.3, the observed response at the cutoff cannot be attributed to a single mechanism under this design. The effect could reflect mechanical amplification of pre-existing age-related performance differences, a behavioral response to the new within-school incentive, or some combination. The empirical results below are therefore interpreted as the reduced-form effect of the reform on the age-cutoff discontinuity, without a commitment to a single channel.

4.3 Specification Choices

Our baseline specification follows the recommendations of Cattaneo et al. (2020) for regression discontinuity designs. We use a local linear polynomial with MSE-optimal bandwidth selection, a triangular kernel, and heteroskedasticity-consistent (HC1) standard errors. We include a set of predetermined covariates (gender, parental education, family income) to improve precision without affecting the bias-corrected point estimates (Calonico et al., 2014). As robustness checks, we consider alternative polynomial orders, bandwidth selection, and placebo cutoffs at non-July dates. These results are reported in Appendix Section F.

5 Results

5.1 The Reform Widened the Application Gap

The ranking reform widened the application gap at the July 1 cutoff. Table 2 presents RD estimates at the July 1 cutoff for the pre-reform and post-reform periods, along with the difference-in-discontinuities for each outcome. Figure 2 displays these estimates visually.

In the pre-reform period, students on the older side of the cutoff were slightly more likely to apply to college than those on the younger side, though the difference is small and not statistically significant ($\hat{\beta}_{pre} = 0.006$, $p = 0.61$). The post-reform period tells a different story. The application gap at the cutoff increased in magnitude and became highly significant ($\hat{\beta}_{post} = 0.044$, $p < 0.001$). The difference between these two estimates, $\hat{\delta} = 0.038$ ($p = 0.032$), indicates that the reform increased the application gap by 3.8 percentage points, equivalent to a 12 percent increase relative to the pre-reform application rate of 31 percent near the cutoff.

Unconditional enrollment shows a positive but insignificant RD in both periods ($\hat{\beta}_{pre} = 0.014$, $\hat{\beta}_{post} = 0.011$), with a difference-in-discontinuities close to zero ($\hat{\delta} = -0.003$, $p = 0.85$). The reform induced more applications from older students, but these additional applications did not translate into more enrollments on net.

Conditional enrollment among applicants moves in the opposite direction. The pre-reform RD is positive but insignificant ($\hat{\beta}_{pre} = 0.033$), while the post-reform estimate turns negative and marginally significant ($\hat{\beta}_{post} = -0.037$, $p = 0.066$). The difference-in-discontinuities is $\hat{\delta} = -0.070$ ($p = 0.032$), a decline of roughly 13 percent relative to a baseline conditional enrollment rate of 56 percent. The reform brought new applicants to the centralized system, but these marginal applicants were less likely to convert their applications into enrollments. We explore plausible mechanisms behind this margin in Section 5.3.

To inspect the temporal composition of these estimates, Appendix E reports a year-by-year decomposition of each outcome. The pre-reform RDs are individually and jointly indistinguishable from zero across the three main outcomes, consistent with the identifying assumption that the July 1 discontinuity was stable in the years before the reform. Within the post-reform window, the response

is concentrated in the 2015 cohort. Two non-mutually-exclusive factors plausibly contribute to this timing. First, programs appear to have taken time to settle into stable ranking weights after the reform: 2013 was a transitional year at the 10 percent regulatory floor, 2014 weights stabilized further, and by 2015 the average ranking weight exceeded the floor (Appendix Figure A.6). Second, families and students may have taken time to learn the new incentive structure and to adjust their application behavior accordingly.

The next subsections investigate whether these results are driven by compositional changes at the threshold, what plausible mechanisms might underlie the decline in conditional enrollment, and whether the application-gap response varies across socioeconomic groups.

5.2 Compositional Changes Do Not Drive the Result

For the application and enrollment results to be attributed to the reform, it must have changed the behavior of students near the July 1 cutoff without changing who those students are. If the composition of students near the threshold shifted differentially after the reform, the observed increase in applications could partly reflect selection into the sample rather than a response to the admissions formula.

Figure 3 and Appendix Table A.1 present difference-in-discontinuities estimates for predetermined covariates and academic performance measures. The post-reform period shows some directional patterns in the individual RD estimates. Older students near the cutoff tend to come from slightly higher-income families, and their average PSU scores are marginally higher than in the pre-reform period. These patterns, however, are present in the pre-reform period as well, and the difference-in-discontinuities for each predetermined covariate (gender, parental education, family income, average PSU score) is statistically indistinguishable from zero, with p -values ranging from 0.11 to 0.84. A joint test of the null that all five predetermined covariate differences are simultaneously zero fails to reject ($\chi^2(5) = 5.58$, $p = 0.35$), and individual p -values remain insignificant after Bonferroni correction for multiple testing (minimum adjusted $p = 0.55$). The relevant test for the difference-in-discontinuities design is whether the change in covariate balance coincides with the reform, not whether each period exhibits perfect balance independently (Grembi et al., 2016). Moreover, in our baseline specifica-

tion, we control for these covariates, absorbing any smooth compositional differences at the threshold (Calónico et al., 2014).

Concha-Arriagada (2023) documents that the ranking reform induced strategic school switching among 12th-grade students seeking to improve their within-school rank. While this behavioral response changed the composition of students at some schools, there is no reason to believe it would differentially affect students born on opposite sides of the July 1 cutoff within the same school, as the switching decision depends on GPA relative to school peers rather than on birth timing.

GPA, unlike the predetermined covariates, is determined during schooling and can respond to the reform through behavioral channels. The difference-in-discontinuities for GPA is positive and statistically significant at the 5% level ($\hat{\delta} = 0.082$, $p = 0.037$), indicating that, after the reform, students on the older side of the cutoff earned higher grades than those on the younger side. A GPA is in part a school-level assessment of its own students, and schools can adjust how they grade in response to changes in admissions incentives. Fajnzylber et al. (2019) document that Chilean schools inflated grades following the ranking reform. The post-reform widening at the cutoff is therefore consistent with a range of plausible channels, including school-level responses to the new incentive, individual effort responses, and a mechanical amplification of pre-existing differences. The GPA shift does not threaten identification because GPA is downstream of the cutoff and is not predetermined at birth, and as noted in Section 2.3 the present design cannot separate these channels.

5.3 Why Does Conditional Enrollment Fall?

The negative conditional enrollment result could arise through several channels. Using the detailed application records from DEMRE, we can investigate two of them. Marginal applicants drawn into the centralized system by the reform may target programs beyond their competitive reach and fail to gain admission. Alternatively, students who receive an offer may choose not to enroll, opting instead for alternatives outside the centralized system, such as vocational programs, non-CRUCH universities, or direct entry into the labor force. We refer to these channels as overreach and deterrence, respectively.

To distinguish them, we decompose the conditional enrollment margin into an admission component and a post-admission component. For each applicant, we observe whether the centralized

matching algorithm selected them for at least one program in their preference list, and whether they ultimately enrolled. Figure 5 displays the difference-in-discontinuities estimates for each decomposition outcome, Appendix Figure A.7 shows the underlying pre- and post-reform RD coefficients, and Appendix Table A.6 reports the full estimates.

The pre-reform RD for admission to any listed program is slightly positive ($\hat{\beta}_{pre} = 0.022$), indicating that older applicants were somewhat more likely to be admitted before the reform. After the reform, this estimate turns negative ($\hat{\beta}_{post} = -0.024$), yielding a difference-in-discontinuities of -0.046 ($p = 0.103$). The pattern is similar for first-choice admission, where the post-reform estimate shifts from near zero to -0.055 , with a difference-in-discontinuities of -0.054 ($p = 0.102$). These estimates sharpen when we narrow the bandwidth to focus on students closest to the cutoff (CER-optimal), with first-choice admission falling by 6.4 percentage points ($p < 0.10$). The quadratic specification attenuates both estimates toward zero, consistent with a local effect concentrated near the threshold.

The deterrence channel also shows directional evidence. Among admitted students, the probability of not enrolling shifted from $\hat{\beta}_{pre} = -0.018$ to $\hat{\beta}_{post} = 0.028$, with a difference-in-discontinuities of $+0.046$ ($p = 0.159$). The sign is stable across bandwidth choices but does not reach conventional significance.

Neither channel individually accounts for the full decline in conditional enrollment. Both point in the expected direction, and together they are consistent with the reform drawing marginal applicants who both aim higher than their competitive position warrants and are somewhat more likely to pursue alternatives when they do receive an offer. This pattern parallels the broader finding that marginal applicants induced into the college pipeline by information or cost changes often face barriers downstream (Hoxby and Avery, 2012; Pallais, 2015).

5.4 Which Families Respond to the Reform?

The application-gap response at the cutoff is not uniform across socioeconomic groups. Table 3 and Figure 4 present difference-in-discontinuities estimates for college application across six cells defined by parental education (primary, secondary, college) and family income (low, high).

Panel A of Table 3 shows that students from primary-educated families exhibit positive but in-

significant effects in both income groups ($\hat{\delta} = 0.041$ for low income, $\hat{\delta} = 0.032$ for high income). The formula change alone appears insufficient to overcome the larger barriers facing families where neither parent completed secondary school.

Panel B reveals where the reform effect concentrates. Among secondary-educated families, the low-income group shows a positive but insignificant estimate ($\hat{\delta} = 0.043$, $p = 0.22$). The high-income group, however, produces the largest and most precisely estimated effect in the table ($\hat{\delta} = 0.113$, $p = 0.006$). The pre-reform RD for this cell is slightly negative ($\hat{\beta}_{pre} = -0.036$), meaning that before the reform, older students from these families were not more likely to apply. After the reform, the estimate becomes strongly positive ($\hat{\beta}_{post} = 0.077$, $p = 0.004$). This sign reversal drives the difference-in-discontinuities.

Panel C confirms the null result for college-educated families. Both income groups produce estimates close to zero ($\hat{\delta} = -0.002$ for low income, $\hat{\delta} = -0.039$ for high income). These families' children were already likely to apply regardless of the ranking reform. This null finding is consistent with [Zimmerman \(2019\)](#), who documents that in the Chilean context, the returns to elite college admission accrue primarily to students from advantaged backgrounds who were already on the margin of admission.

This pattern is consistent with the reform creating a new incentive along the within-school relative-performance margin, with the response concentrated among families positioned to decode and act on it. Findings from rank-based admissions policies in other contexts document similar heterogeneity ([Cullen et al., 2013](#); [Niu and Tienda, 2010](#)). Students from secondary-educated, higher-income families occupy a specific position in the college access pipeline. Their parents did not attend college, so the student would be a first-generation applicant, but the family has sufficient resources to support the application process. For this group, the reform plausibly shifted the perceived return to applying, either by changing admissions scores directly or by prompting behavioral responses that raised those scores.

Figure 4 displays the difference-in-discontinuities estimates for each education-income category. The visual pattern reinforces the statistical result: only the secondary-educated, higher-income cell produces a precisely estimated positive effect, while all other cells straddle zero.

6 Conclusion

Admissions formula design can produce differential effects along demographic dimensions the designer did not target. Chile’s 2013 reform shifted weight within the university admissions formula from absolute GPA toward a within-school class ranking component, raising the return to within-school relative performance. Using a difference-in-discontinuities design, this paper shows that the reform widened the application gap at the July 1 school entry cutoff by 3.8 percentage points (12 percent). The gap in GPA at the same cutoff also emerged after the reform ($\hat{\delta} = 0.082$, $p = 0.037$). The cutoff, which before the reform produced no pooled gap in applications, now produces a large one.

These additional applications did not translate into additional enrollments. Conditional enrollment among applicants fell by 7.0 percentage points, with patterns consistent with marginal applicants who both aim higher than their competitive position warrants and are somewhat more likely to pursue alternatives when admitted. The response concentrates among students from secondary-educated, higher-income families ($\hat{\delta} = 0.113$, $p = 0.006$), with no effect among college-educated families whose children apply regardless or among primary-educated families facing larger barriers. The formula change matters most for families at the margin of college participation, where it shifts the perceived return to applying.

Three implications follow. First, admissions formulas are not neutral instruments. The ranking reform was designed to reward students who excel relative to their school context, yet it produced differential effects at a demographic cutoff the designer did not target. [Concha-Arriagada \(2023\)](#) documents a parallel consequence of the same reform, finding that strategic school switching eroded approximately 30 percent of the intended redistribution. Formulas based on within-school relative performance appear vulnerable to responses that can undermine their equity objectives. Second, the application margin deserves attention as a distinct outcome in studies of college access. The three-margin decomposition reveals a large effect on applications but no net effect on enrollment, a finding invisible in enrollment-only analyses. Third, admissions reforms are most consequential not for the poorest or the most advantaged families, but for those in between, where a formula change alters the perceived competitiveness of their application.

Several limitations apply. Our sample conditions on PSU test-taking, so we do not observe whether

the reform changed who takes the entrance exam. [Celhay and Gallegos \(2025\)](#) document a positive age-at-entry effect on PSU take-up for overlapping Chilean cohorts, suggesting our estimates capture the intensive margin among test-takers rather than the full effect. The on-schedule restriction (Section 3.1) further narrows the sample and introduces selection that the present design cannot address. Because we estimate intention-to-treat effects, our estimates reflect the reduced-form policy impact of the cutoff rather than the per-year effect of delayed enrollment. The three-year window (2010–2015) avoids contamination from Chile’s 2016 free tuition reform but limits our ability to study whether the effect persists as schools and families adapt to the ranking component. The design also does not cleanly separate the channels through which the reform produced the observed cutoff response: mechanical amplification of pre-existing academic differences, school-level grading responses to the ranking incentive, and individual effort responses are all compatible with the estimated patterns. Finally, the paper studies how the reform affected students near the age cutoff, but not the infra-marginal students who would apply regardless. Whether the reform improved program match quality for always-applicants, by giving them higher scores that unlock more selective programs, is an important direction for future work.

Table 1: Summary Statistics Around the July 1 Cutoff

	Pre-reform			Post-reform		
	Younger	Older	Diff	Younger	Older	Diff
Panel A: Outcome variables						
Applied	0.309	0.332	0.023	0.372	0.404	0.032
Enrolled	0.173	0.188	0.014	0.235	0.259	0.024
GPA score	-0.021	0.033	0.054	-0.030	0.045	0.076
Ranking score	—	—	—	-0.030	0.048	0.078
Panel B: Baseline covariates						
Male	0.474	0.485	0.011	0.471	0.476	0.005
Income: middle	0.311	0.313	0.002	0.392	0.397	0.004
Income: high	0.120	0.126	0.006	0.153	0.160	0.008
Educ: secondary	0.482	0.488	0.006	0.505	0.501	-0.004
Educ: college	0.230	0.234	0.004	0.251	0.257	0.006
<i>N</i>	23,457	25,428		22,999	24,819	

Note: Sample consists of on-schedule PSU first-takers in the July window born within ± 15 days of the July 1 cutoff. Younger = born before July 1; Older = born on or after July 1. Diff = Older – Younger (unadjusted means). Ranking score shown for post-reform cohorts only (official DEMRE data). The difference-in-discontinuities estimate is recovered via `rdrobust` in Table 2.

Table 2: Effect of the Ranking Reform on College Outcomes

	(1)	(2)	(3)
	$\hat{\beta}_{pre}$	$\hat{\beta}_{post}$	(2) – (1)
Applied	0.006 (0.012)	0.044*** (0.013)	0.038** (0.018)
Enrolled	0.014 (0.010)	0.011 (0.011)	-0.003 (0.015)
Enrolled (if applied)	0.033 (0.024)	-0.037* (0.022)	-0.070** (0.032)
Covariates	Yes	Yes	
Bandwidth (days)	24	22	
N	300,201	294,671	

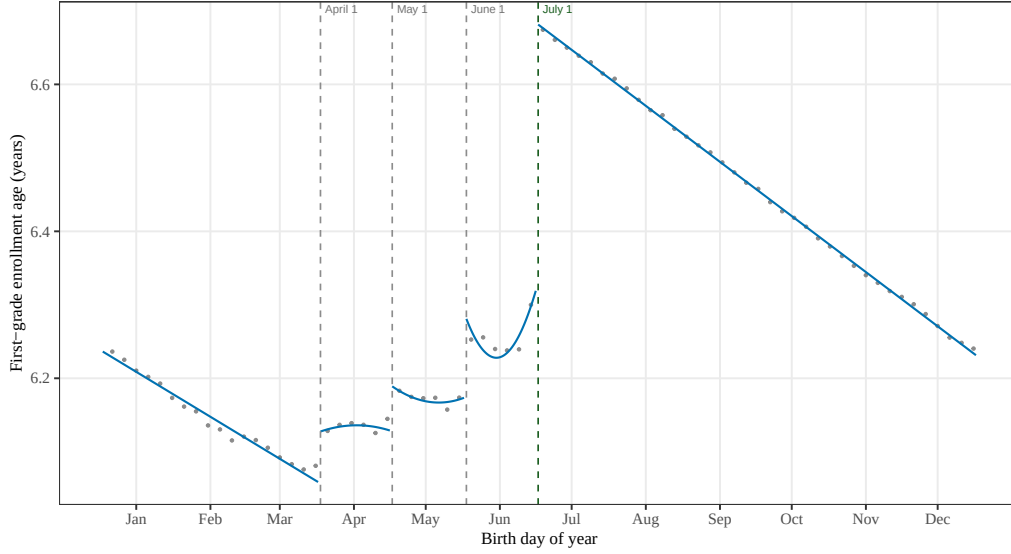
Note: Each row reports RD estimates at the July 1 school entry cutoff for the indicated outcome. Column (1) reports the pre-reform estimate, column (2) the post-reform estimate, and column (3) the difference-in-discontinuities. Enrolled (if applied) is estimated on the applicant subsample. Pre-reform means (younger group, within bandwidth): Applied = 0.309, Enrolled = 0.173, Enrolled (if applied) = 0.558. Covariates include gender, family income, and parental education. Estimator: bias-corrected local polynomial with robust standard errors (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Heterogeneous Reform Effects on College Application by Family Background

	(1)	(2)	(3)
	$\hat{\beta}_{pre}$	$\hat{\beta}_{post}$	(2) – (1)
Panel A: Primary educ. parents			
Low income	-0.019 (0.020)	0.023 (0.025)	0.041 (0.032)
High income	0.062 (0.052)	0.094** (0.045)	0.032 (0.069)
Panel B: Secondary educ. parents			
Low income	0.022 (0.023)	0.065** (0.026)	0.043 (0.035)
High income	-0.036 (0.031)	0.077*** (0.027)	0.113*** (0.041)
Panel C: College educ. parents			
Low income	-0.012 (0.066)	-0.014 (0.075)	-0.002 (0.100)
High income	0.046 (0.032)	0.008 (0.029)	-0.039 (0.043)

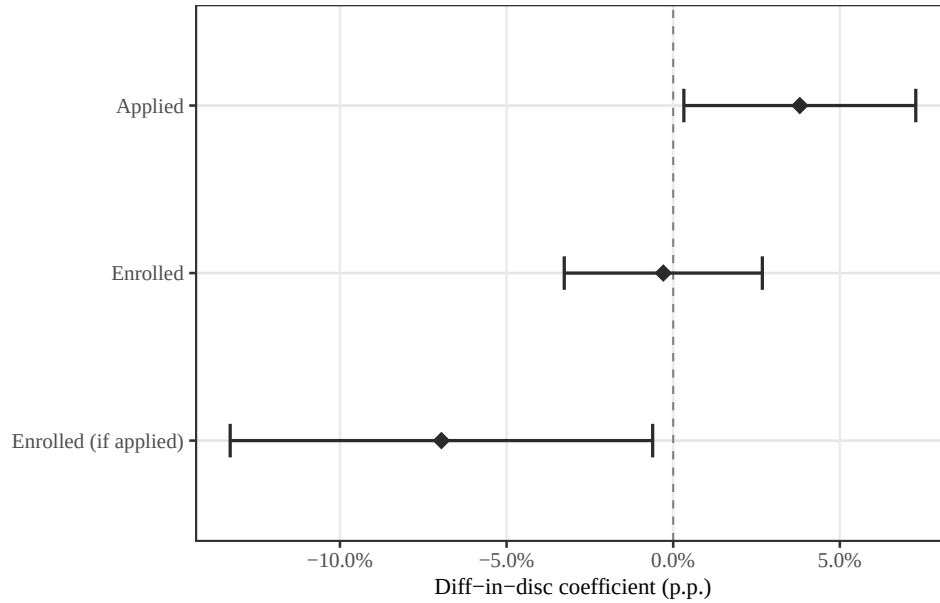
Note: Each row reports difference-in-discontinuities estimates for college application (=1) within the indicated education and income cell. Column (1) reports the pre-reform RD, column (2) the post-reform RD, and column (3) the difference-in-discontinuities. Education groups are mutually exclusive (highest parental attainment). Income is based on self-reported household brackets from the DEMRE survey. Cell sample sizes range from 10,674 to 84,646 (within MSE-optimal bandwidth). Covariate: gender (education and income dummies excluded within each cell). Estimation follows (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1: First-stage: school starting age by day of birth



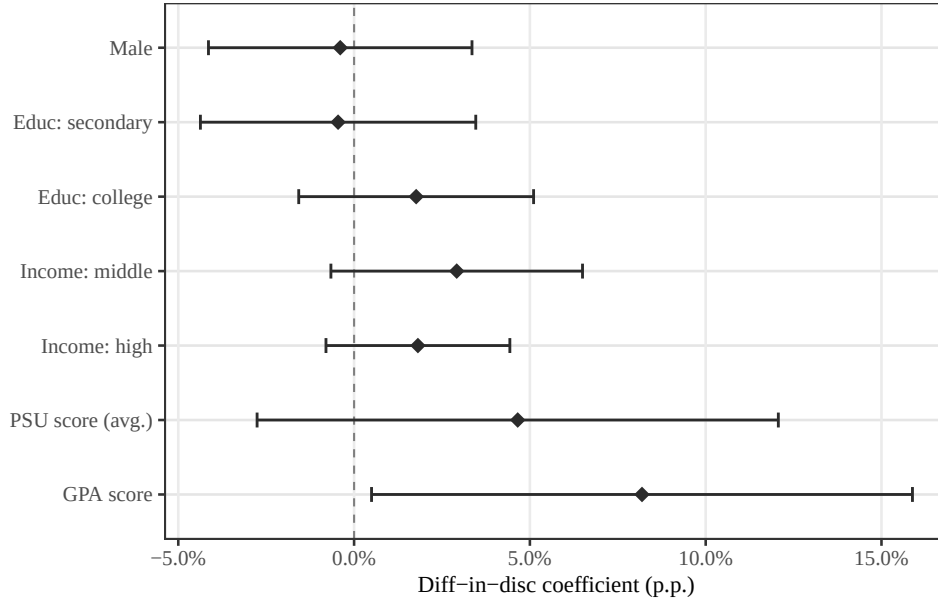
Note: This figure plots the relationship between day of birth in the calendar year and first-grade enrollment age. Each point represents the average enrollment age within a day-of-birth bin. Vertical lines mark the four school entry cutoffs: April 1, May 1, June 1, and July 1. The July 1 cutoff produces the largest discontinuity (approximately 0.4 years).

Figure 2: Difference-in-discontinuities estimates for main outcomes



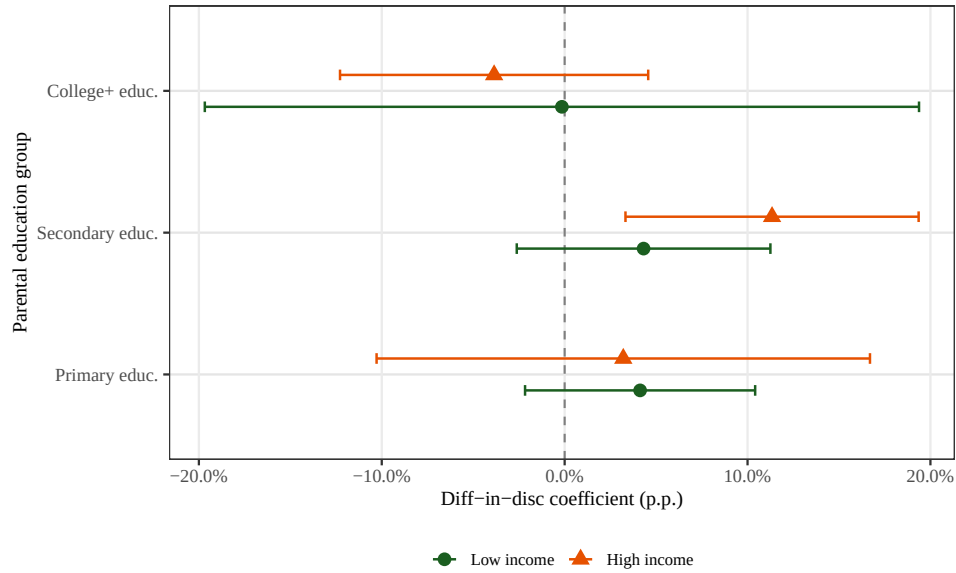
Note: This figure plots DiDisc estimates at the July 1 cutoff for each main outcome. College application and unconditional enrollment are estimated on the full sample of PSU takers. Conditional enrollment is estimated on the applicant subsample. Horizontal bars represent 95% confidence intervals. RD estimation follows [Calonico et al. \(2014\)](#), difference-in-discontinuities design follows [Grembi et al. \(2016\)](#).

Figure 3: Covariate balance at the July 1 cutoff



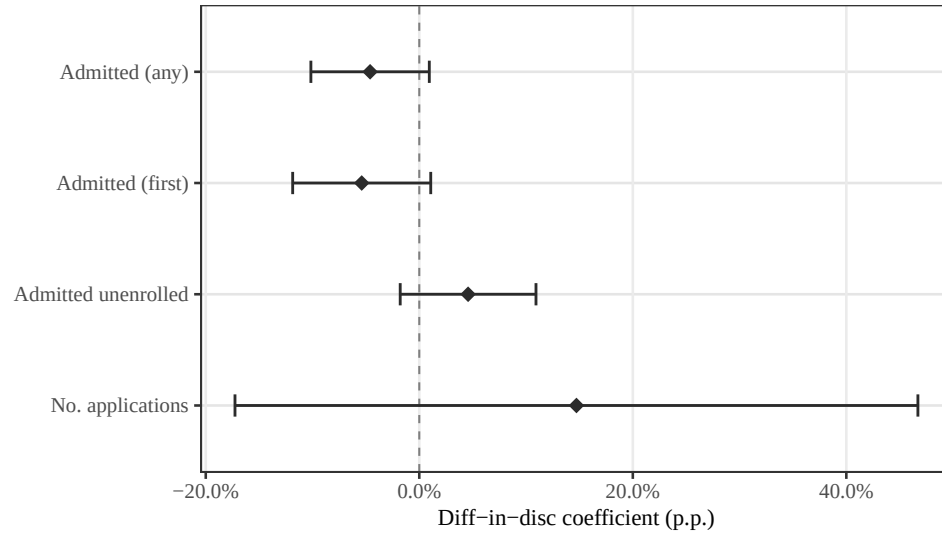
Note: This figure plots DiDisc estimates at the July 1 cutoff for each predetermined covariate. Covariates include gender, parental education (secondary, college), family income (middle, high), and normalized PSU scores (math, verbal). Horizontal bars represent 95% confidence intervals. RD estimation follows [Calonico et al. \(2014\)](#), difference-in-discontinuities design follows [Grembi et al. \(2016\)](#).

Figure 4: Heterogeneous reform effects on college application by education and income



Note: This figure plots DiDisc estimates for college application across six cells defined by parental education (primary, secondary, college) and family income (low, high). Vertical bars represent 95% confidence intervals. RD estimation follows [Calonico et al. \(2014\)](#), difference-in-discontinuities design follows [Grembi et al. \(2016\)](#).

Figure 5: Decomposition of conditional enrollment



Note: This figure plots DiDisc estimates for four outcomes measured among applicants. Horizontal bars represent 95% confidence intervals. RD estimation follows [Calonico et al. \(2014\)](#), difference-in-discontinuities design follows [Grembi et al. \(2016\)](#).

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A Cutoff Selection and Enrollment Age Construction

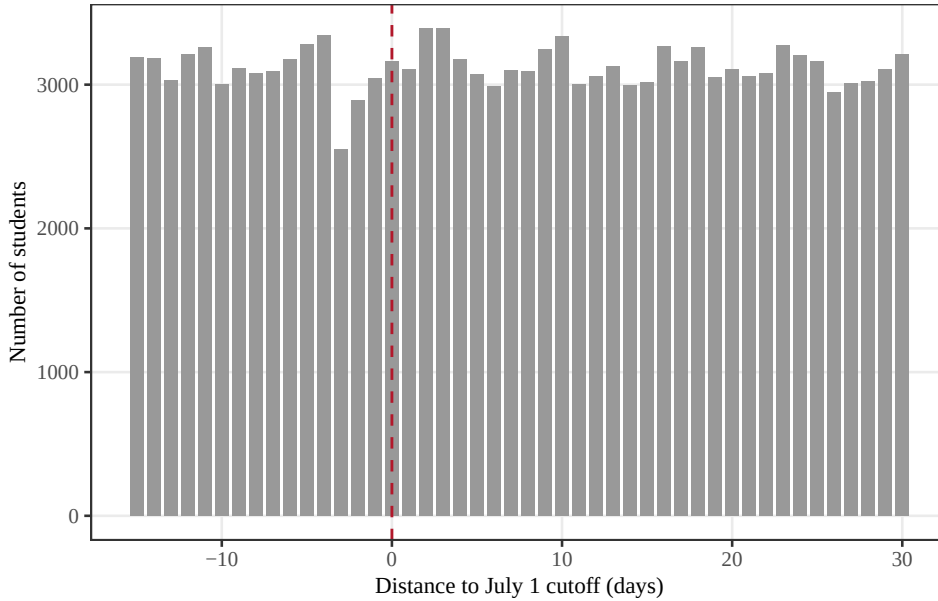
Chile operates four school entry cutoffs (April 1, May 1, June 1, and July 1), each generating a discontinuity in first-grade enrollment age. We focus on the July 1 cutoff because it produces the largest first-stage jump (approximately 0.4 years), reflecting the fact that most schools adopted this date as their operational entry rule. The April, May, and June cutoffs produce substantially smaller jumps (approximately 0.06, 0.06, and 0.11 years, respectively). A Cochran Q test of equality across the four cutoff-specific RD estimates rejects the null of homogeneity for college application ($Q = 18.2$, $df = 3$, $p < 0.001$), confirming that the July 1 effect is statistically distinct.

The enrollment age measure used in Figure 1 combines two sources. For approximately 27 percent of students in our sample (52 percent in the post-reform period), we directly observe the year of first-grade enrollment from the MINEDUC *Rendimiento* panel. For the remainder, we approximate school starting age as the student’s graduation year minus eleven, correcting for observed grade repetitions following [McEwan and Shapiro \(2008\)](#). We validate this approximation against the directly observed measure on the overlap subsample ($r = 0.79$).

Following [Celhay and Gallegos \(2025\)](#), we estimate intention-to-treat effects throughout the paper and do not scale estimates by enrollment age. Doing so would require monotonicity and exclusion restriction assumptions that are difficult to defend when multiple cutoffs generate heterogeneous compliance rates.

B Density Test

Figure A.1: Density of births around the July 1 cutoff



Note: This figure plots the number of students by day of birth relative to the July 1 school entry cutoff. The dashed vertical line marks the cutoff. No visual evidence of bunching is detected ([McCrory, 2008](#)).

C Covariate Balance

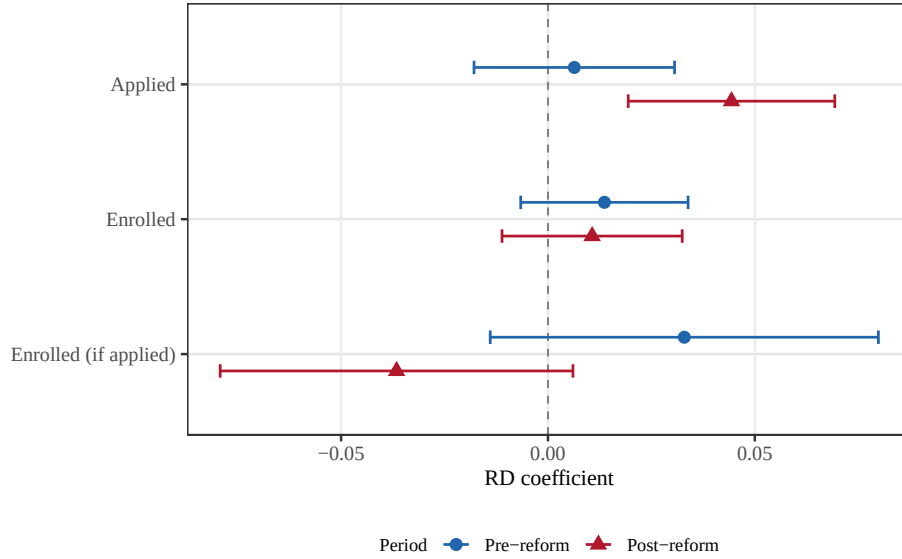
Table A.1: Covariate Balance at the July 1 Cutoff

	(1) $\hat{\beta}_{pre}$	(2) $\hat{\beta}_{post}$	(3) (2) – (1)
Panel A: Predetermined covariates			
Male	0.004 (0.013)	-0.000 (0.014)	-0.004 (0.019)
Educ: secondary	0.015 (0.014)	0.010 (0.014)	-0.005 (0.020)
Educ: college	-0.003 (0.012)	0.014 (0.012)	0.018 (0.017)
Income: middle	-0.013 (0.012)	0.016 (0.013)	0.029 (0.018)
Income: high	0.000 (0.009)	0.018* (0.010)	0.018 (0.013)
<i>Joint significance test</i>			
$\chi^2(5)$			5.58
<i>p</i> -value			0.350
Panel B: Downstream academic measures			
PSU score (avg.)	0.010 (0.026)	0.057** (0.027)	0.047 (0.038)
GPA score	-0.011 (0.027)	0.071** (0.028)	0.082** (0.039)

Note: Each row reports RD estimates at the July 1 cutoff. Column (1): pre-reform period. Column (2): post-reform period. Column (3): difference-in-discontinuities. Panel A reports predetermined covariates (determined before the student enters school). Panel B reports downstream academic measures that can respond to the reform. The joint χ^2 test evaluates whether all Panel A differences are simultaneously zero. Bonferroni-adjusted minimum *p*-value across Panel A: 0.55. Estimation follows (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

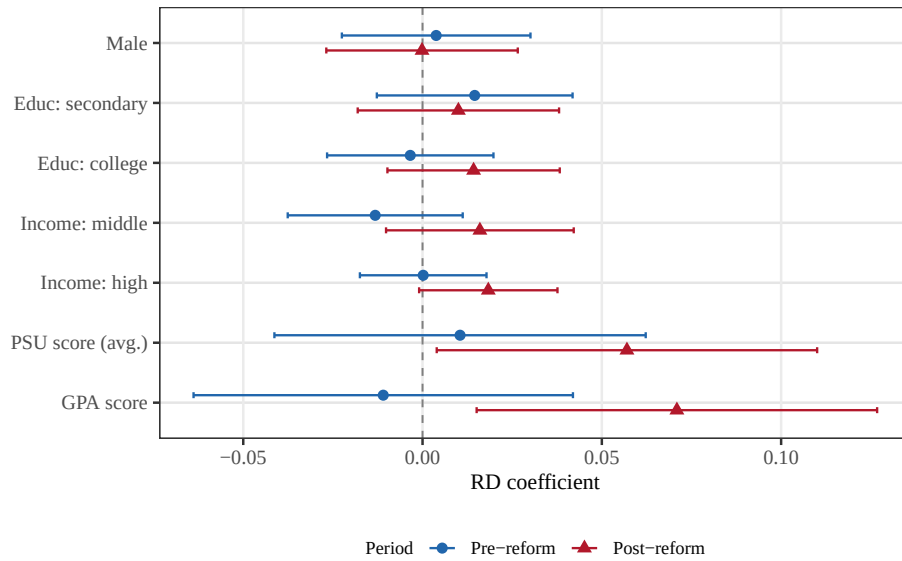
D Pre-post RD Coefficients Plots

Figure A.2: Pre-reform and post-reform RD estimates for main outcomes



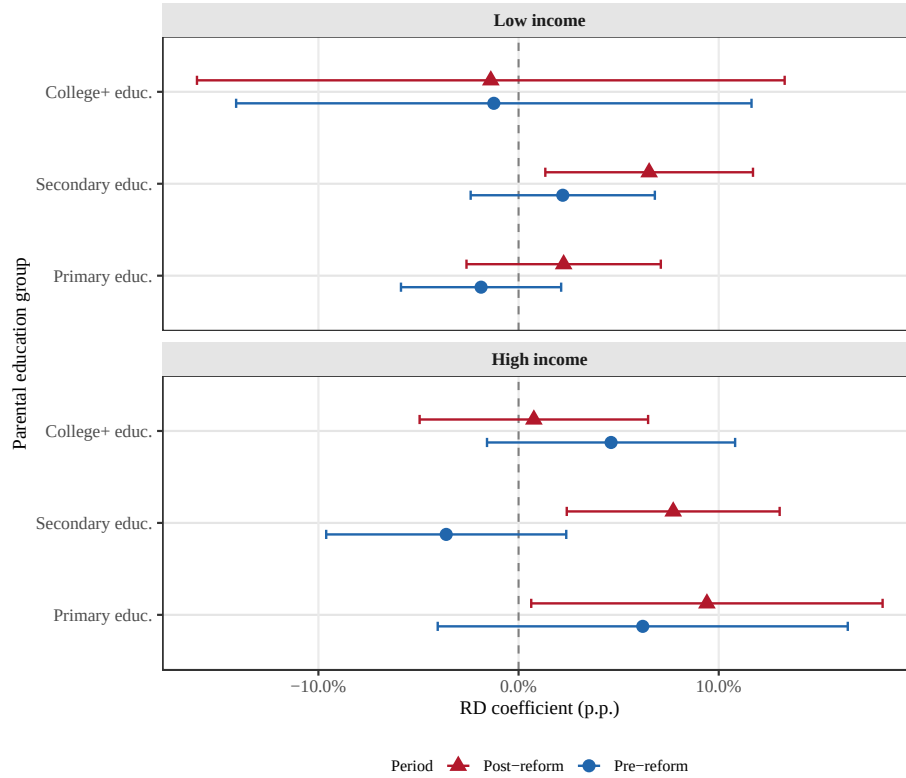
Note: This figure plots RD estimates at the July 1 cutoff for each main outcome, separately for the pre-reform and post-reform periods. College application and unconditional enrollment are estimated on the full sample of PSU takers. Conditional enrollment is estimated on the applicant subsample. Horizontal bars represent 95% confidence intervals. Estimation follows [Calonico et al. \(2014\)](#).

Figure A.3: Covariate balance at the July 1 cutoff



Note: This figure plots RD estimates at the July 1 cutoff for each predetermined covariate, separately for the pre-reform and post-reform periods. Covariates include gender, parental education (secondary, college), family income (middle, high), and normalized PSU scores (math, verbal). Horizontal bars represent 95% confidence intervals. Estimation follows [Calonico et al. \(2014\)](#).

Figure A.4: Heterogeneous reform effects on college application by education and income



Note: This figure plots difference-in-discontinuities estimates for college application across six cells defined by parental education (primary, secondary, college) and family income (low, high). Vertical bars represent 95% confidence intervals. RD estimation follows [Calónico et al. \(2014\)](#), difference-in-discontinuities design follows [Grembi et al. \(2016\)](#).

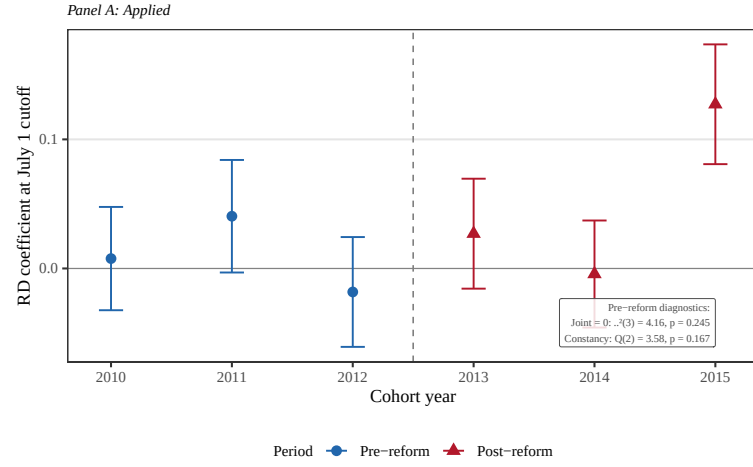
E Year-by-Year Event Study

Section 5 reports pooled pre-reform (2010–2012) and post-reform (2013–2015) difference-in-discontinuities estimates. This appendix decomposes the pooled estimates into cohort-year-specific RD coefficients at the July 1 cutoff, both to test the pre-reform validity of the difference-in-discontinuities design and to document the temporal distribution of the post-reform effect.

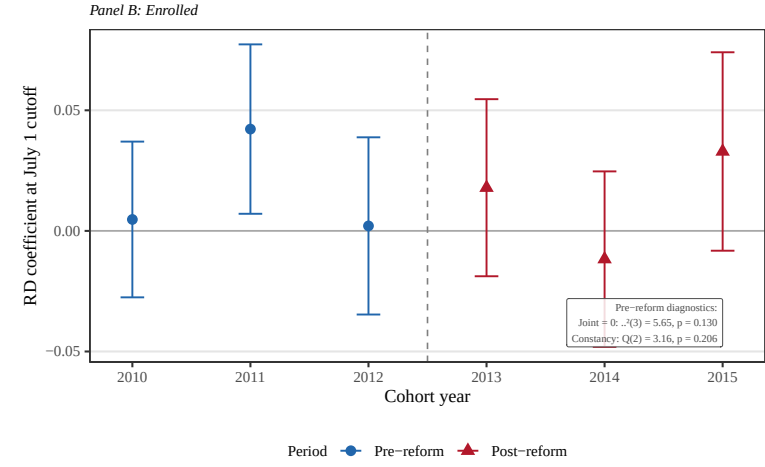
The pre-reform RDs are individually and jointly indistinguishable from zero for all three main outcomes. The joint-zero test fails to reject at conventional levels for college application ($\chi^2(3) = 4.16$, $p = 0.245$), unconditional enrollment ($\chi^2(3) = 5.65$, $p = 0.130$), and conditional enrollment ($\chi^2(3) = 2.99$, $p = 0.393$). The constancy tests also pass in all three cases, supporting the assumption that the magnitude of the July 1 discontinuity was stable in the years before the reform.

The post-reform pattern is more concentrated than the pooled estimates suggest. For college application and conditional enrollment, the statistically significant coefficients in 2013–2015 come from the 2015 cohort alone. The 2013 and 2014 coefficients are individually insignificant and small in magnitude. Two interpretations are consistent with this pattern. First, a behavioral response to the ranking incentive may take time to materialize as students, families, and schools adjust to the new weight structure. Second, programs increased their ranking weights above the 10 percent regulatory floor over the post-reform period (Appendix G), so the intensity of treatment grew across cohorts. The present design cannot distinguish a genuine learning lag from 2015-specific factors, and we flag this as a direction for future work.

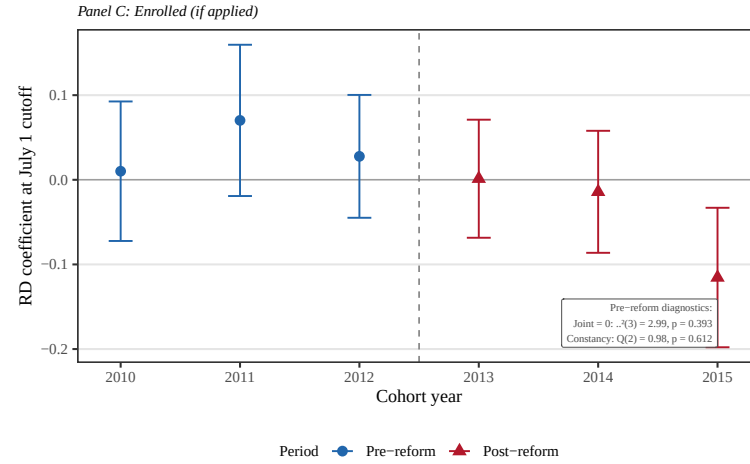
Figure A.5: Year-by-year RD estimates for main outcomes, 2010–2015



(a) Applied



(b) Enrolled



(c) Enrolled (if applied)

Note: Each panel plots RD coefficients at the July 1 cutoff estimated separately by cohort year. Blue points (circles) are pre-reform cohorts (2010–2012); red points (triangles) are post-reform cohorts (2013–2015). Vertical bars are 95% confidence intervals. The dashed vertical line marks the reform. Joint-zero and constancy tests for the pre-reform window are annotated in each panel. Estimation follows [Calonico et al. \(2014\)](#) with MSE-optimal bandwidth, triangular kernel, and HC1 standard errors.

F Robustness

We subject the main difference-in-discontinuities estimates to four robustness checks. The results confirm that the application effect is robust to specification choices while the conditional enrollment effect is equally stable.

Polynomial order and bandwidth selection. Our baseline uses a local linear polynomial ($p = 1$) with MSE-optimal bandwidth. Table A.2 presents estimates under two alternative specifications: a local quadratic ($p = 2$) with MSE-optimal bandwidth, which uses a wider bandwidth and provides a more conservative test. The application estimate is 3.8 percentage points under the baseline and 2.2 percentage points under the quadratic. The attenuation under the quadratic is expected, as broader bandwidths include observations farther from the cutoff, diluting a treatment effect that is concentrated near the threshold. The conditional enrollment result follows the same pattern, with the linear specification producing a significant estimate (-7.0 pp) and the quadratic attenuating toward zero.

Table A.2: Robustness: Polynomial Order

	(1) $\hat{\beta}_{pre}$	(2) $\hat{\beta}_{post}$	(3) (2) – (1)
Panel A: Applied			
$p = 1$ (bw: 24 / 22)	0.006 (0.012)	0.044*** (0.013)	0.038** (0.018)
$p = 2$ (bw: 27 / 26)	0.014 (0.018)	0.036* (0.019)	0.022 (0.026)

Note: Each row reports difference-in-discontinuities estimates for college application under an alternative polynomial order. Bandwidth (bw) in days shown per specification. Covariates include gender, family income, and parental education. Estimation follows (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Alternative fixed bandwidths. Table A.3 reports estimates for college application at bandwidths of $0.5h$, $0.75h$, h (MSE-optimal), $1.5h$, and $2.0h$ days around the July 1 cutoff. The point estimate is stable across this range, and the confidence intervals overlap throughout. Narrower bandwidths produce larger point estimates with wider confidence intervals, consistent with a local effect.

Placebo cutoffs. If the result reflects the ranking reform rather than a generic change at the July 1 discontinuity, we should observe no differential response at dates where no school entry rule applies. Table A.4 tests three placebo cutoffs (October 1, May 15, and March 1) and confirms this prediction. The estimates are small, statistically insignificant, and of mixed sign. The response is specific to the July 1 cutoff.

No covariates. Our baseline specification includes a set of predetermined covariates (gender, parental education, family income) to improve precision. Removing all covariates yields a difference-in-discontinuities of 4.5 percentage points for college application ($p = 0.014$), slightly larger than the baseline estimate, confirming that covariate adjustment does not drive the result.

Table A.3: Robustness: Bandwidth Sensitivity

	(1) $\hat{\beta}_{pre}$	(2) $\hat{\beta}_{post}$	(3) (2) – (1)
Panel A: Applied			
0.50×	0.007 (0.016)	0.052*** (0.018)	0.045* (0.024)
0.75×	0.010 (0.013)	0.044*** (0.014)	0.034* (0.019)
1.0× (baseline)	0.007 (0.013)	0.046*** (0.013)	0.039** (0.018)
1.50×	0.005 (0.012)	0.045*** (0.012)	0.039** (0.017)
2.00×	0.002 (0.012)	0.045*** (0.012)	0.043** (0.017)

Note: Each row reports difference-in-discontinuities estimates for college application at alternative bandwidths expressed as multiples of the MSE-optimal bandwidth ($h_{pre} = 24$, $h_{post} = 22$ days). Covariates include gender, family income, and parental education. Estimation follows (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Robustness: Placebo Cutoffs

	(1) $\hat{\beta}_{pre}$	(2) $\hat{\beta}_{post}$	(3) (2) – (1)
Panel A: True cutoff (July 1)			
Baseline	0.006 (0.012)	0.044*** (0.013)	0.038** (0.018)
Panel B: Placebo cutoffs			
Oct 1 (day 274)	-0.014 (0.012)	-0.017* (0.010)	-0.003 (0.015)
May 15 (day 135)	-0.004 (0.010)	0.002 (0.011)	0.006 (0.014)
Mar 1 (day 60)	0.005 (0.011)	-0.008 (0.010)	-0.013 (0.015)

Note: Panel A reproduces the baseline estimate at the July 1 cutoff. Panel B reports estimates at placebo cutoff dates where no school entry rule applies. Placebo windows span ± 60 days around each date. Covariates include gender, family income, and parental education. Estimation follows (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

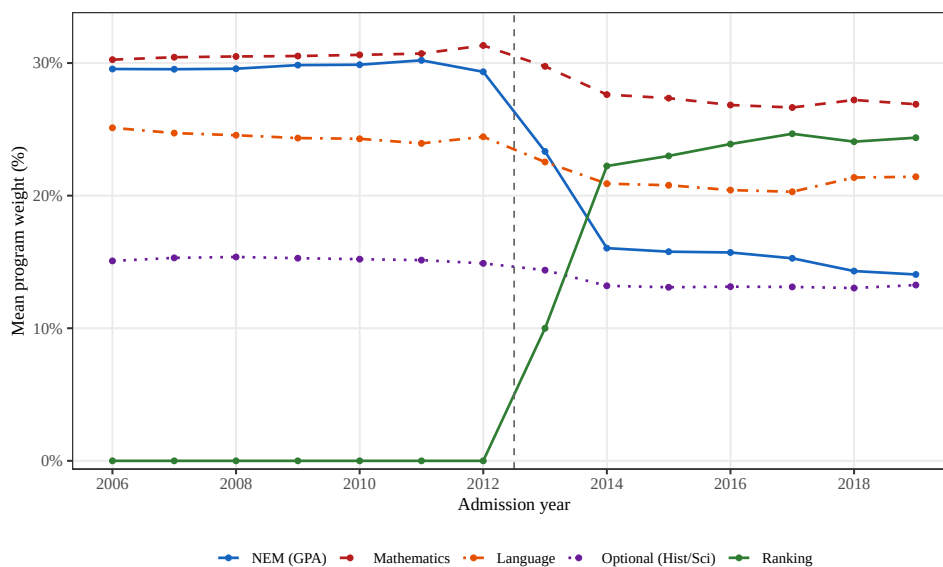
G Program-Level Weight Shift

Table A.5: Admissions Weight Composition Before and After the Reform

	Pre-reform	Post-reform	Change
NEM (GPA)	29.7	16.2	-13.5
Mathematics	30.7	27.4	-3.2
Language	24.5	21.1	-3.4
Optional (Hist/Sci)	15.2	13.3	-1.9
Ranking (new)	0.0	22.0	+22.0
Total	100.0	100.0	
N (program $\times$ year)	6,988	10,638	

Note: Mean weight assigned to each admissions score component across all programs in the centralized system. Weights are on a 0–100 percentage point scale and sum to approximately 100. The ranking score was introduced in 2013, funded by reductions in other components. Optional is the maximum of history and science weights per program. Estimation follows (Calonico et al., 2014).

Figure A.6: Admissions weight composition before and after the reform



Note: This figure plots the average weight assigned to each admissions score component across all programs in the centralized system. The ranking score was introduced in 2013, funded primarily by reductions in NEM weight. Five components sum to approximately 100 percentage points.

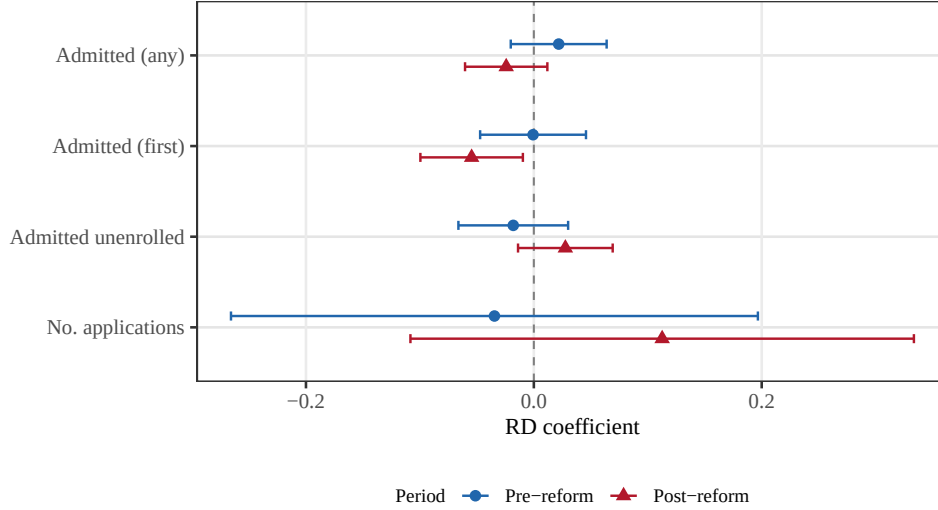
H Enrollment Decomposition

Table A.6: Decomposition of Conditional Enrollment

	(1) $\hat{\beta}_{pre}$	(2) $\hat{\beta}_{post}$	(3) (2) – (1)
Admitted (any)	0.022 (0.021)	-0.024 (0.018)	-0.046 (0.028)
Admitted (first)	-0.001 (0.024)	-0.055** (0.023)	-0.054 (0.033)
Admitted unenrolled	-0.018 (0.025)	0.028 (0.021)	0.046 (0.032)
No. applications	-0.035 (0.118)	0.113 (0.113)	0.147 (0.163)
N	98,264	116,977	

Note: Sample consists of applicants (PSU takers who submitted at least one preference). Admitted (any) indicates selection by the DAA for at least one listed program. Admitted (1st choice) indicates selection for the top-listed preference. Admitted unenrolled indicates selection but no matriculation in any centralized program. No. preferences is the count of programs listed in the application. Estimation follows (Calonico et al., 2014). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

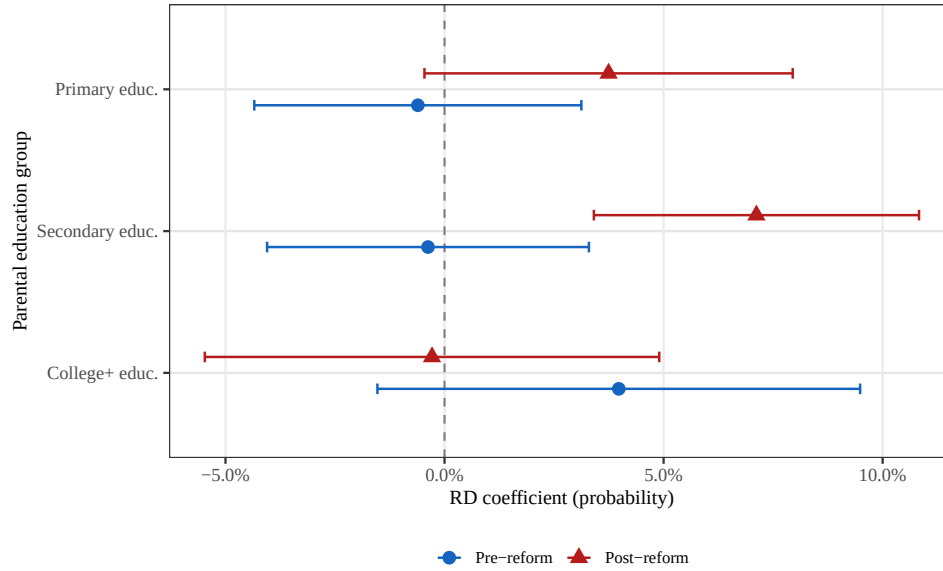
Figure A.7: Decomposition of conditional enrollment



Note: This figure plots pre-reform and post-reform RD coefficients at the July 1 cutoff for four outcomes measured among applicants. Horizontal bars represent 95% confidence intervals. Estimation follows Calonico et al. (2014).

I Additional SES Specifications

Figure A.8: Pre-reform and post-reform RD estimates by parental education



Note: This figure plots the pre-reform and post-reform RD coefficients at the July 1 cutoff for college application, separately by parental education group (primary, secondary, college). Vertical bars represent 95% confidence intervals. Estimation follows [Calonico et al. \(2014\)](#).